

Effect of the Intermolecular Hydrogen Bond between Carbazole and *N,N*-Dimethylformamide/Isopropanolamine on the Solubility of Carbazole

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ABSTRACT: The influence of hydrogen bonding on the solubility of carbazole and anthracene in *N,N*-dimethylformamide (DMF) and isopropanolamine (IPA) is investigated accordingly by ^1H nuclear magnetic resonance (NMR) analysis. The chemical shift for the free and hydrogen-bonding proton of the anthracene and carbazole solution in DMF, IPA, and the mixture of the two was collected by a 600 MHz ^1H NMR spectrometer. It is proposed that DMF, IPA, and a mixture of the two would be able to efficiently refine carbazole from crude anthracene oil. This phenomenon is shown to be the result of the intermolecular hydrogen bond of $\text{N}-\text{H}\cdots\text{O}$ and $\text{N}-\text{H}\cdots\text{N}$ formed between carbazole and the DMF/IPA mixture solvent, which greatly enhanced the solubility of carbazole in DMF and IPA. The arising steric hindrance effects derived from the intermolecular hydrogen bonding between DMF and IPA result in the significant solubility decline of anthracene and carbazole in the DMF and IPA mixture.

1. INTRODUCTION

Under the conditions of today's oil shortage, coal tar from the coal carbonization process is an extremely valuable primary fuel. On the one hand, it can produce diesel and fuel oil. Meanwhile, coal tar is aromatic hydrocarbon-based organic mixtures. Among them, anthracene (AN), phenanthrene, and carbazole (CAR) are the main components. These aromatic hydrocarbon compounds being irreplaceable high-valued products in petrochemical products could be processed by hydrogenation and solvent extraction.

Separation of coal derivatives is the important aspect of coal use. CAR and its derivatives, primarily refined from the crude AN of coal tar¹ and partly extracted from petroleum crude oil,² are of great importance to the chemical industry as intermediates in dyes, pigments, medicine, pesticides, and photoelectric materials.^{2–4} The key to the separation process is how to select a suitable solvent system. Thus far, to understand the interaction between solvent and solute, it is still a "like dissolves like" rule. The solvent method is one of the most effective processes in preparing high-purity CAR from crude AN oil. The selection of the solvent system used is principally based on the solubility of AN and CAR in different solvents. However, there is a lack of data regarding the solubility and solid–liquid equilibrium of CAR and AN in different solvents because of the time-consuming and work-intensive nature of the processes.^{5,6} Therefore, researchers have more recently tended to focus on the solubility determination and prediction by empirical or semi-empirical activity coefficient models.^{7–10} Nevertheless, the resulting predictions have Hobson's choice and tremendously restrict flexibility in the selection of solvents, because activity coefficient models are often developed on the basis of liquid–liquid or vapor–liquid equilibrium theory. The solubility selectivity of solvent on CAR and AN is the key basis of the solvent selection used in the separation process of AN

and CAR. The investigation of solvent–solute interactions is one of the most effective approaches to achieve the selection of solvents with high solubility selectivity on CAR and AN.^{2,11} The interaction of solute and solvent generally includes van der Waals forces and hydrogen bonds. Hydrogen bonds exist widely in solution systems,^{12,13} polymers,¹⁴ biological molecules,^{15,16} and crystals.^{17–19} According to Steiner et al.,¹⁶ the energy of π hydrogen bonds range from 8.4 to 17 kJ mol^{−1}. It has also been found that the energy of general hydrogen bonds may range from 4 to 170 kJ mol^{−1}.¹⁴ Generally, the formation of hydrogen bonds can effectively promote the dissolution of the solute, and in some case, alternatively, can also restrain the dissolution of the solute.²⁰ However, little work has been performed to explore the interaction between AN/CAR and solvent. Clearly, to know the intermolecular mechanism between solvent and solute under different circumstances would be a scientific guidance about coal tar refining.

In the AN and CAR molecules, all of the carbon atoms are sp²-hybridized. For the CAR molecules, because of the strong electron-withdrawing ability of aromatic rings, the N–H group tends to more readily form hydrogen bonds with a strong electron donor. In addition, the *p* electron not involved in the hybrid has a higher delocalization energy; therefore, the aromatic rings would act as the electron donor and, thereby, form C–H $\cdots\pi$ hydrogen bonds with *N,N*-dimethylformamide (DMF).²¹

Fourier transform (FT)-Raman,²² infrared (IR), nuclear magnetic resonance (NMR), nuclear quadrupole resonance (NQR) spectroscopy,²³ X-ray diffraction (XRD), and ultra-violet–visible (UV–vis)^{15,24} are often the most useful tools in

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detecting weak hydrogen bonds,^{13,14,16,18,19,25,26} and FT-Raman, IR, and XRD are generally adapted to detect the hydrogen bonds in solids or crystals. The hydrogen bonds may result in a decrease in diamagnetic shielding around the bridging proton, causing the resonance of the proton to shift to a lower field. Conventional hydrogen bonds can lead to a chemical shift, causing the bridging proton to move downfield by 1–4 ppm, and the C–H···O hydrogen bond can also lead to a downfield chemical shift of 0.4–1.0 ppm in the bridging proton.²⁷

It is well-known that DMF is of high solubility selectivity on AN and CAR. To improve the solubility selectivity of DMF, isopropanolamine (IPA), a double functional group solvent, has been added to DMF. In this paper, ¹H NMR spectrometry has been used to detect the hydrogen bond between CAR and DMF and IPA. Solubility data for CAR in DMF, IPA, and the mixed solvent were also determined. The effects of the hydrogen bond between CAR and the solvent (including DMF, IPA, and the mixed solvent) on the solubility of CAR in those solvents are discussed.

2. EXPERIMENTAL SECTION

2.1. Samples and Preparation. AN and CAR were purified by recrystallization with DMF. AN was dissolved at 130 °C and filtered at 50 °C. CAR was dissolved at 110 °C and filtered at 30 °C. AN and CAR purities were analyzed by gas chromatography (GC).

2.2. ¹H NMR Analysis. Excess AN and CAR were dissolved in DMF, IPA, and the mixed solvent at 25 °C. Subsequently, the saturated solutions were diluted by deuteriochloroform containing tetramethylsilane (99:1, v/v).

¹H NMR spectra were recorded on a Bruker AVANCEIII-600 NMR spectrometer at 600 MHz, at 25 °C. Proton chemical shifts were referenced to tetramethylsilane (TMS) in CDCl₃. The pulse program was zg30 1D sequence, using a 30° flip angle. The result is a routine proton NMR spectrum. The data were analyzed using MestRenova software (version 6.1.1-6384). In the ¹H NMR spectra of AN, CAR, DMF, and IPA, all of the proton chemical shifts were assigned using advanced chemistry development (ACD/Laboratories) software, version 6.0.

2.3. Solubility Measurement. The solubility of AN and CAR in DMF, IPA, and the mixed solvent was measured by the equilibrium method. AN and CAR were dissolved in DMF and IPA to the saturation state at 90 °C. The system temperature was then gradually reduced by 10 °C intervals to a final temperature of 30 °C and kept at equilibrium for 3 days. The samples were analyzed by GC with an internal standard method, employing 9-fluorenone as the internal standard compound.

GC analysis was carried out on a Shimadzu GC-2014 instrument, equipped with a hydrogen flame ionization detector (FID). The instrument was operated by manual injection with a 30:1 split ratio, and the injection port temperature was maintained at 280 °C. The injection volume was 0.4 μL. The column was a RTX-5, 30 m × 0.32 mm × 0.25 μm, wall-coated with 5% diphenyl-/95% dimethylpolysiloxane. The oven temperature was held at 170 °C for 1 min, increased to 200 °C at 3.5 °C min⁻¹, kept at 200 °C for 1 min, increased to 250 °C at 10 °C min⁻¹, and held at 250 °C for 1 min. The carrier gas was helium of 99.999% purity with a constant flow rate of 34.0 mL min⁻¹.

The selectivity (the solubility selectivity of solvent to CAR and AN) is given by the following formula:

$$\text{selectivity} = \frac{S_{\text{CAR}}}{S_{\text{AN}}}$$

where S_{CAR} and S_{AN} are the solubilities of CAR and AN in the solvent, respectively.

3. RESULTS AND DISCUSSION

3.1. Solubility of AN and CAR in DMF and IPA. The solubility curves of CAR and AN in DMF and IPA are shown in Figure 1. It is clear that AN has roughly 8 times the solubility

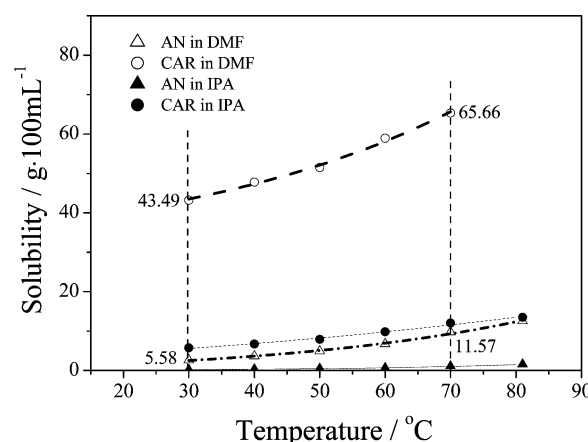


Figure 1. Solubility curve of CAR and AN in DMF and IPA solvent.

than CAR in DMF at 30 °C and 6 times at 70 °C, but AN and CAR both have less solubility in IPA in the temperature range of 30–70 °C, with CAR having a slightly bigger solubility in IPA. The selectivity of DMF and IPA to AN and CAR are shown in Figure 2. It can be observed from Figure 2 that IPA

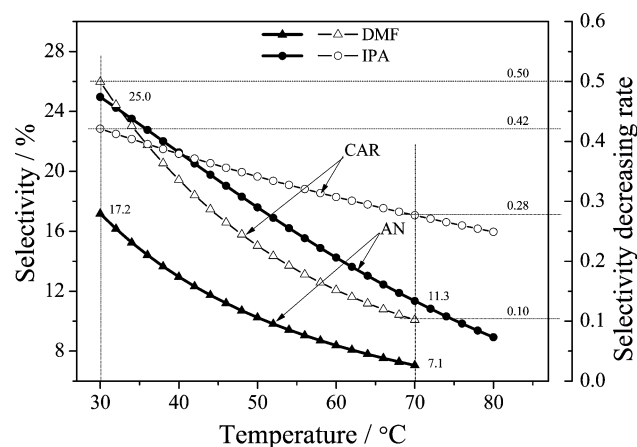


Figure 2. Solubility selectivity of DMF and IPA to AN and CAR.

has a higher selectivity to CAR than DMF and the selectivity of DMF and IPA declines rapidly with an increase in the temperature. At 70 °C, the selectivity of DMF and IPA are 7.1 and 11.3, respectively, but at a lower temperature of 30 °C, the selectivity of DMF and IPA reached 17.2 and 25.0, respectively. The selectivity decreasing rate of DMF is larger than that of IPA at 30 °C, but at 36 °C and above, the selectivity decreasing rate of IPA is larger than that of DMF.

In the DMF/IPA mixed solvent with a 20% volume fraction of IPA, the solubility of AN and CAR are 4.17 and 33.81 g 100 mL⁻¹, respectively, at 70 °C, approximately 0.3 and 0.5 times that in DMF (Figure 3). When the volume fraction of IPA was increased to 35%, the solubility of AN and CAR in the mixed solvent only reached 2.12 and 18.15 g 100 mL⁻¹, respectively, at 70 °C. It can be concluded that the addition of IPA to DMF can significantly inhibit the dissolution of AN and CAR in

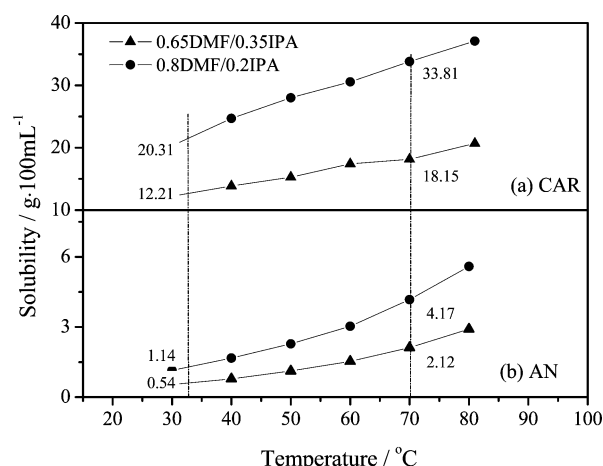


Figure 3. Solubility of AN and CAR in varying volume ratios of the DMF/IPA mixed solvent.

DMF and that the inhibiting effect obviously increases with an increasing volume fraction of IPA in mixed solvent.

3.2. Hydrogen Bond in the AN–DMF/IPA System.

Figure 4 shows the ^1H NMR spectra of AN, DMF, and IPA in deuteriochloroform and AN in deuteriochloroform with DMF/IPA solvent. In deuteriochloroform, the characteristic chemical shifts of AN (Figure 4a) are 7.468, 8.007, and 8.427 ppm. In comparison to panels d and e of Figure 4, the movement of the

chemical shifts for the protons of the AN molecule are not significant, and no new peak appears in the spectra of AN solution in deuteriochloroform containing DMF or IPA solvent. Additionally, in comparison to the proton chemical shift of DMF and IPA (panels d and e of Figure 4), the movement of the chemical shift for DMF and IPA (panels b and c of Figure 4) is negligible. Therefore, the conventional hydrogen bond and π hydrogen bond between AN and solvent DMF and IPA were not detected.

In the characteristic ^1H NMR spectra of IPA, the chemical shift for NH_2 is a broad peak, which is due to the presence of water in deuteriochloroform. The proton chemical exchange between IPA and the trace amount of water in deuteriochloroform lead to the formation of the wide peak of NH_2 . The IPA molecule has two negatively charged centers, the O and N atoms, which are surrounded by one and two protons, respectively. Therefore, IPA may easily provide two proton acceptors and two proton donors, forming intermolecular hydrogen bonds with other IPA molecules. Figure 4c, a detailed view of the chemical shifts, suggests that intermolecular hydrogen bonding between IPA molecules takes place at the OH and NH_2 groups. Because of the electronegativity (EN) difference of O (EN of 3.44) and N (EN of 3.04) atoms, the various forms of intermolecular hydrogen bonds are possible, including $\text{O}-\text{H}\cdots\text{O}$, $\text{O}-\text{H}\cdots\text{N}$ (also $\text{N}-\text{H}\cdots\text{O}$), and $\text{N}-\text{H}\cdots\text{N}$. The corresponding proton chemical shifts are 3.51, 3.24, and 3.01 ppm, respectively.

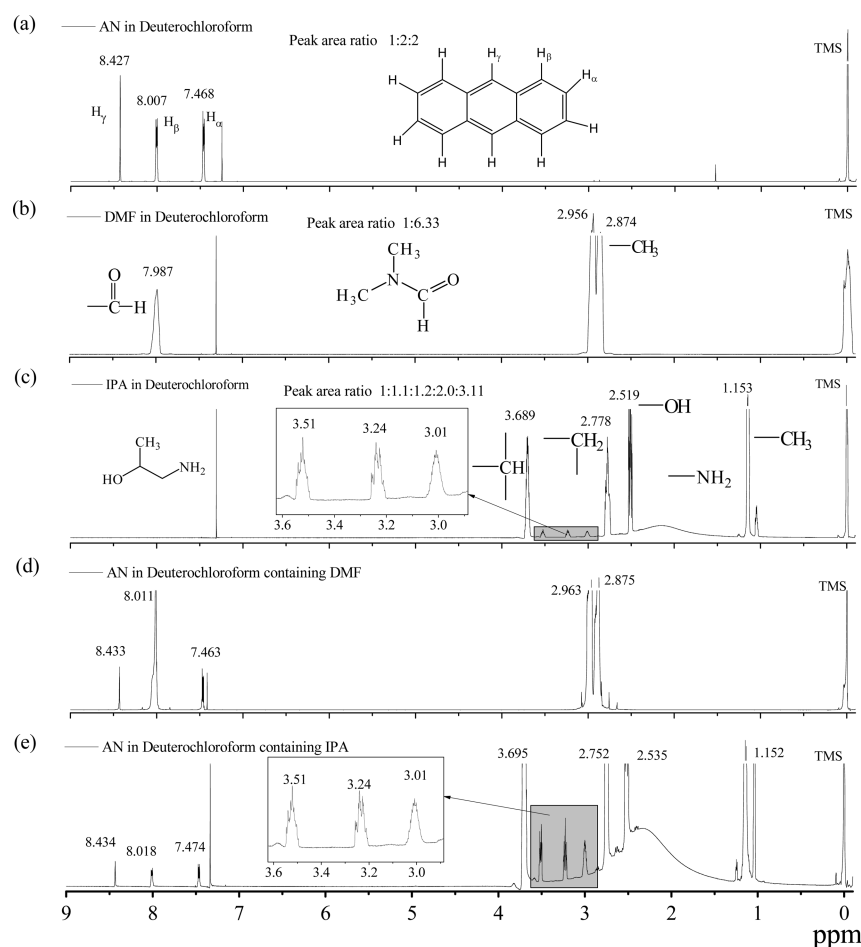


Figure 4. ^1H NMR spectra of AN, DMF, and IPA in deuteriochloroform and AN in deuteriochloroform with DMF/IPA solvent.

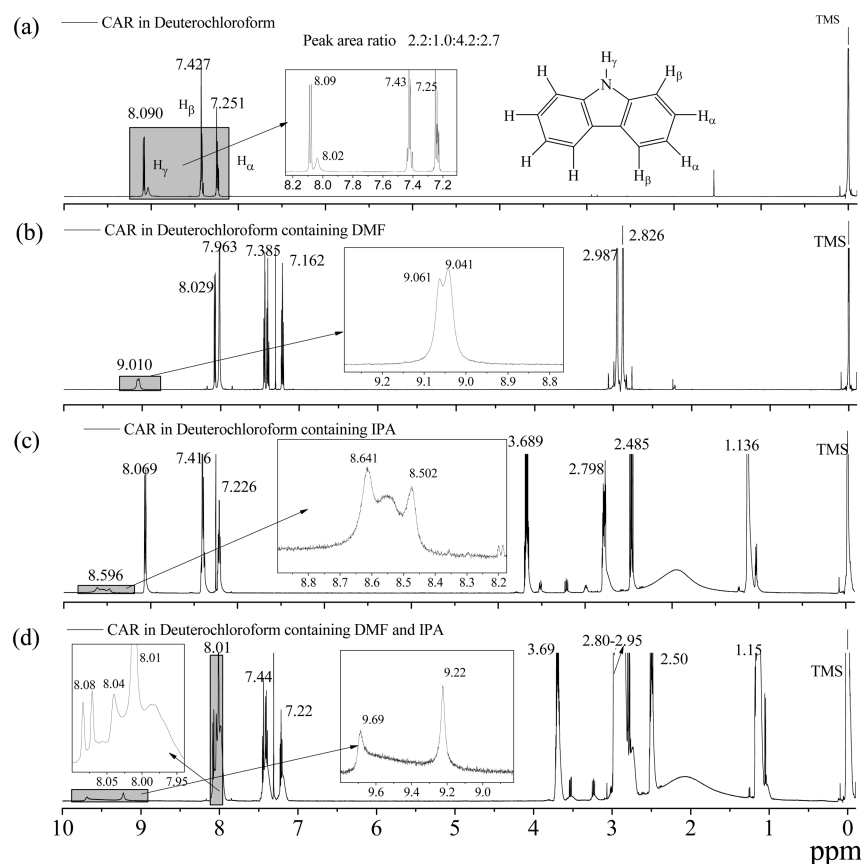


Figure 5. ^1H NMR spectra of CAR in deuteriochloroform with DMF/IPA solvent.

3.3. Hydrogen Bond in the CAR–DMF/IPA System.

Figure 5 presents the ^1H NMR spectra of CAR in deuteriochloroform with DMF/IPA solvent. In Figure 5a, the characteristic chemical shifts of CAR are 7.25, 7.43, 8.02, and 8.09 ppm. Integral results show that the peak area ratio is approximately 2:1:4:3, not coinciding with the number of protons of the CAR molecule. In accordance with research by Kitawaki et al.,²⁸ it is probable that, in the N-substituted CAR derivatives, all of the protons in 9-carbazoyl exist in three different states, despite the various effects of different substituents, and the chemical shifts are 7.24, 7.37–7.45, and 8.12 ppm. In comparison to the ^1H NMR spectra of CAR in deuteriochloroform, the chemical shift for H_γ of the CAR molecule is 8.02–8.09 ppm.

All of the free and hydrogen-bonding proton peak areas of CAR in the four systems were integrated and presented in Table 1.

Table 1. Proton Peak Area of CAR in Deuteriochloroform Containing DMF/IPA Solvent

chemical shift (ppm)	proton peak area of CAR in deuteriochloroform			
	deuteriochloroform	DMF	IPA	DMF/IPA
7.24	1.24	0.99	1.00	1.03
7.43	1.96	1.98	2.04	2.08
8.02	0.42			0.26
8.02–8.09	1.00	1.00	1.00	1.00
9.04–9.06		0.43		
8.50–8.64			0.40	
9.22–9.69				0.14

The results presented in Table 1 suggest that, in the CAR solution in deuteriochloroform containing DMF and IPA, the chemical shift for H_γ of the CAR molecule moves from 8.02 to 9.04–9.06 and 8.50–8.64 ppm, respectively. When CAR was dissolved in the deuteriochloroform containing the DMF–IPA mixture, the chemical shifts for part of H_γ moved from 8.02 to 9.22 and 9.69 ppm. The new chemical shifts of 9.06, 8.50–8.64, and 9.22–9.69 ppm imply an increase in the deshielding effect of H_γ in CAR. It is believed that the intermolecular hydrogen bonds were formed between CAR and the solvent DMF and IPA molecule, resulting in a higher chemical shift for H_γ . On the basis of the peak area data in Table 1, it can be concluded that almost all of the CAR molecules in the solvent DMF and IPA participated in the formation of intermolecular hydrogen bonds, while only 30% CAR molecules from the CAR solution in the DMF/IPA mixed solvent participated in the formation of intermolecular hydrogen bonds. In comparison to DMF and IPA, the mixed solvent has resulted in a larger chemical shift variation for H_γ of the CAR molecule.

In the DMF molecule shown in Figure 6, two negatively charged centers are provided by O and N; however, N is surrounded by two methyl groups. Because of the steric hindrance caused by the presence of the two methyl groups, hydrogen bonding between N and CAR is much less favorable than between C=O and CAR, as ^1H NMR spectra results have shown. In the spectra analysis, only one new peak appears at 9.06 ppm. The N–H group (positively charged center) of CAR acts as a proton donor, while the O of the C=O group (negatively charged center) of DMF acts as a proton acceptor.¹² Similarly, C=O...H–N hydrogen bonding was reported between cyanuric acid and pyrrole by Sehnert et al.²⁹

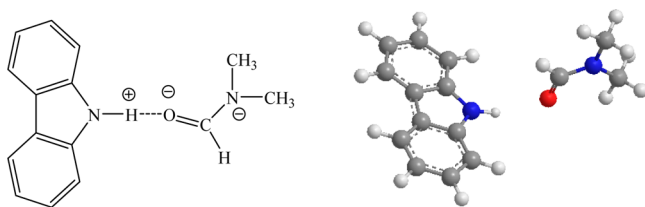


Figure 6. Schematic diagram and geometric structure of the intermolecular hydrogen bond in the CAR–DMF system. Color coding: red, O atom; blue, N atom; gray, C atom; and white, H atom.

In the spectra analysis from Figure 5c, the two peaks that appear at 8.50 and 8.64 ppm suggest that N–H···O and N–H···N intermolecular hydrogen bonds occur between CAR and IPA. The IPA molecule contains two negatively charged centers, provided by the OH and NH₂ groups, which can act as either the donor or acceptor.³⁰ As a result of higher electron density and electronegativity of O than N, the N–H···O and N–H···N intermolecular hydrogen bonds between CAR and IPA result in a chemical shift movement for H_γ of CAR from 8.02 to 8.50–8.64 ppm, respectively, as shown in Figure 7.

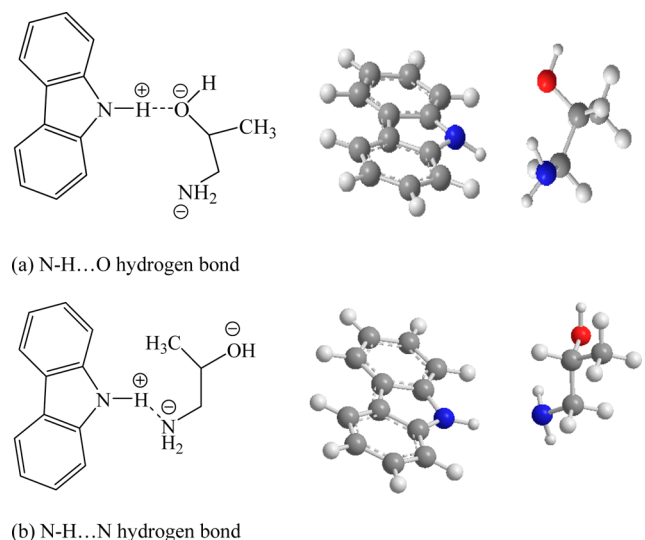


Figure 7. Schematic diagram and geometric structure of the intermolecular hydrogen bond in the CAR–IPA system. Color coding: red, O atom; blue, N atom; gray, C atom; and white, H atom.

It is generally accepted that the chemical shift of the hydrogen-bond proton can be significantly affected by the solute concentration.²⁷ A higher solute concentration would lead to the hydrogen-bond proton causing a much larger chemical shift. In the DMF/IPA mixed solvent, the chemical shift of the hydrogen-bond proton H_γ is much greater than that found with either DMF or IPA (Figure 5 and Table 1), even though the solubility of CAR in the DMF/IPA mixed solvent is far below that of DMF.

This could be explained by the fact that DMF and IPA are both low-molecular-weight components, readily forming intermolecular hydrogen bonds. Furthermore, in the DMF/IPA mixed solvent, the intermolecular hydrogen bonds formed between DMF and IPA greatly reduce the interaction sites of DMF with CAR, effectively inhibiting the dissolution of CAR. Dependent upon the orientation of hydrogen bonds, it is possible for O in DMF to simultaneously form hydrogen bonds

in two orientations with CAR and IPA, resulting in complex hydrogen-bond network structures, as shown in Figure 8.

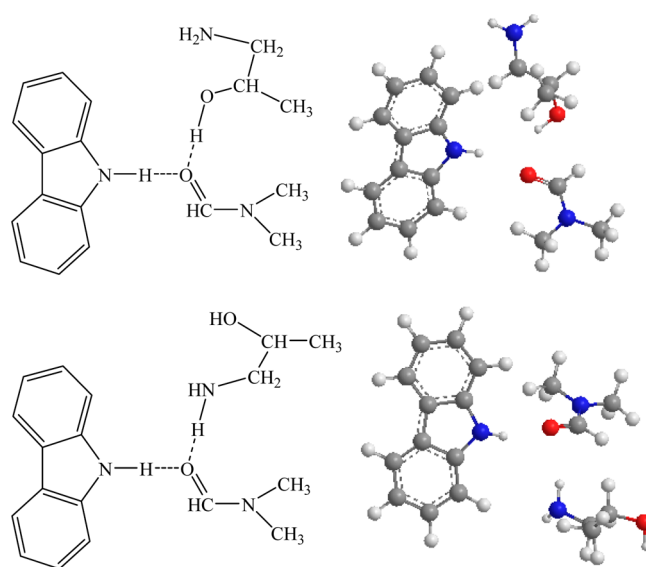


Figure 8. Hydrogen-bond network among CAR, DMF, and IPA. Color coding: red, O atom; blue, N atom; gray, C atom; and white, H atom.

DMF and IPA would be acting as the proton acceptor and donor, respectively. The formation of hydrogen bonds is an electron cloud density equalization process for the atoms participating in the hydrogen bonding in the static electric field, and the electron cloud density of the proton acceptor and donor of the negatively charged atoms decreases. Because of the electron cloud density decrease of O of the O=C group of DMF by the influence of the hydrogen bonding between DMF and IPA, the electron cloud density of the hydrogen-bonding proton H_γ partly moves to O of O=C, further strengthening the deshielding effect of H_γ and resulting in a much higher chemical shift for the hydrogen-bonding proton H_γ of CAR. These results are similar to those found in the investigation of the influence of hydrogen bonding on C=O and O–H functional groups in carboxylic solids using ¹⁷O NMR by Wong et al.³¹ The experimental and calculated results of ¹⁷O NMR by Wong et al. in Table 2 indicated that the formation of intermolecular hydrogen bonding between acrylic acid results in a low isotropic chemical shift (δ_{iso}) for the C=O group. Similar conclusions were drawn by Yamada et al.³² The CAR–

Table 2. Calculated Isotropic Chemical Shifts for C=O and O–H Oxygen in an Intermolecular Hydrogen-Bonding Model of Acrylic Acid³¹

$r(\text{C}=\text{O} \cdots \text{H}-\text{O})$ (Å)	$r(\text{O} \cdots \text{O})$ (Å)	δ_{iso} for C=O oxygen ^a (ppm)
1.433	2.383	320.8
1.568	2.518	325.9
1.703	2.652	330.9
2.150	3.100	344.2
2.874	3.824	357.1
monomer		370.1

^a $\delta_{\text{iso}} = \sigma_{\text{ref}} - \sigma_{\text{iso}}$, where σ_{ref} and σ_{iso} are the magnetic shielding constants for the reference and object, respectively.

DMF–IPA hydrogen-bonding networks are presented in Figure 8.

3.4. Effect of the Hydrogen Bond on the Solubility of CAR in the DMF/IPA Mixture. On the basis of the solubility curves of AN and CAR and solubility selectivity curves of DMF and IPA to CAR and AN in Figures 1–3, in temperatures ranging from 30 to 80 °C, the solubility of CAR is shown to be much higher than that of AN in solvent solution. The selectivity of the investigated solvents decreased significantly with an increase in the temperature, and the selectivity decreasing rate of DMF and IPA dropped from 0.5 and 0.42 to 0.1 and 0.28, respectively. The temperature has a strong influence on the hydrogen bonding, and elongation of the bond length even terminates with an increase in the temperature. Therefore, the decreasing solubility selectivity with the increase in the temperature has a close relationship with the hydrogen bond between CAR and DMF/IPA solvents.

In comparison to the hydrogen-bond formation between CAR and solvents, the hydrogen bonding between DMF and IPA in a mixed solvent solution greatly constrains the solubility of AN and CAR. The formation of hydrogen bonds between DMF and IPA contributes greatly to the steric hindrance, while at the same time, the participation of IPA consumes possible CAR–DMF interaction sites. Steric hindrance from the formation of hydrogen bonds between DMF and IPA plays the key role in limiting the dissolution of AN and CAR in the DMF/IPA mixed solvent.

4. CONCLUSION

The ^1H NMR results show that strong hydrogen bonds are readily formed between CAR and DMF/IPA at 25 °C but that hydrogen bonding between AN and DMF/IPA is less probable. On the basis of the ^1H NMR results, it was found that only the conventional hydrogen bonds formed between CAR and DMF or IPA, while complex hydrogen-bond networks formed between CAR and the DMF/IPA mixed solvent. The $\text{C}=\text{O}\cdots\text{H}-\text{N}$, $\text{N}\cdots\text{H}-\text{N}$, and $\text{O}\cdots\text{H}-\text{N}$ hydrogen bonds are the significance factor in the high solubility of CAR in DMF, IPA, and the mixed solvent and high selectivity to CAR at lower temperatures. However, the formation of intermolecular hydrogen bonds between DMF and IPA limits the solubility of AN and CAR in the DMF/IPA mixed solvent because of the steric hindrance from the hydrogen bonding of DMF and IPA. Therefore, the hydrogen-bond interaction can be regarded as the key factor of the selection of solvent being suitable for the separation of AN and CAR. In addition, the solvent with two functional groups (OH and NH_2) acts only as a proton acceptor, and it will be a recommendation for future solvent selection.

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Notes

The authors declare no competing financial interest.

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